

## Working with Selenium 3

automationexercise.com

**Step 1:** create a maven project on terminal by typing the command **mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenAutomationExercise -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false**

```
samiksha@samiksha:~/Devops60/MyMavenPracticeAutomationApp$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenExerciseAutomationApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.456 s
[INFO] Finished at: 2026-03-17T20:40:07+05:30
[INFO] -----
```

**Step 2 :** edit the pom.xml file and add the dependencies

The pom.xml has the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
```

```
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenExerciseAutomationApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenExerciseAutomationApp</name>
  <url>http://maven.apache.org</url>
```

```
<dependencies>
```

```
  <dependency>
```

```
    <groupId>junit</groupId>
```

```
    <artifactId>junit</artifactId>
```

```
    <version>3.8.1</version>
```

```
    <scope>test</scope>
```

```
  </dependency>
```

```
<!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
```

```
<dependency>
```

```
  <groupId>org.seleniumhq.selenium</groupId>
```

```
  <artifactId>selenium-java</artifactId>
```

```
  <version>4.29.0</version>
```

```
</dependency>
```

```
<dependency>
```

```
  <groupId>io.github.bonigarcia</groupId>
```

```
  <artifactId>webdrivermanager</artifactId>
```

```

        <version>5.8.0</version>
    </dependency>
</dependencies>
<!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
    <groupId>org.apache.maven.plugins</groupId>

    <artifactId>maven-jar-plugin</artifactId>
    <version>3.0.1</version>
    <configuration>
        <archive>
            <manifestEntries>
                <Main-Class>com.example.App</Main-Class>
            </manifestEntries>
        </archive>
    </configuration>
</plugin>
</plugins>
</build>
</project>

```

**Step 3:** Edit the App.java

package com.example;

import org.openqa.selenium.By;

```

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.JavascriptExecutor;
import org.openqa.selenium.support.ui.WebDriverWait;
import org.openqa.selenium.support.ui.ExpectedConditions;

import io.github.bonigarcia.wdm.WebDriverManager;
import java.time.Duration;

public class App {
    public static void main(String[] args) {

        WebDriverManager.chromedriver().setup();
        WebDriver driver = new ChromeDriver();
        driver.manage().window().maximize();

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));

        try {
            driver.get("https://automationexercise.com");

            // Click Products page

            wait.until(ExpectedConditions.elementToBeClickable(By.xpath("//a[@href='/products']"
            ))).click();

            // Find first Add to Cart button
            WebElement addToCart = wait.until(ExpectedConditions.elementToBeClickable(
                By.xpath("//a[@data-product-id='1']")));

            // Scroll and click using JavaScript (avoids ad overlay)
            ((JavascriptExecutor) driver).executeScript("arguments[0].scrollIntoView(true);",
            addToCart);
            ((JavascriptExecutor) driver).executeScript("arguments[0].click();", addToCart);

            // Wait for confirmation modal
            wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("cartModal")));

            // Click View Cart
            wait.until(ExpectedConditions.elementToBeClickable(By.linkText("View
            Cart"))).click();

            System.out.println("Product successfully added to cart!");
        }
    }
}

```

```

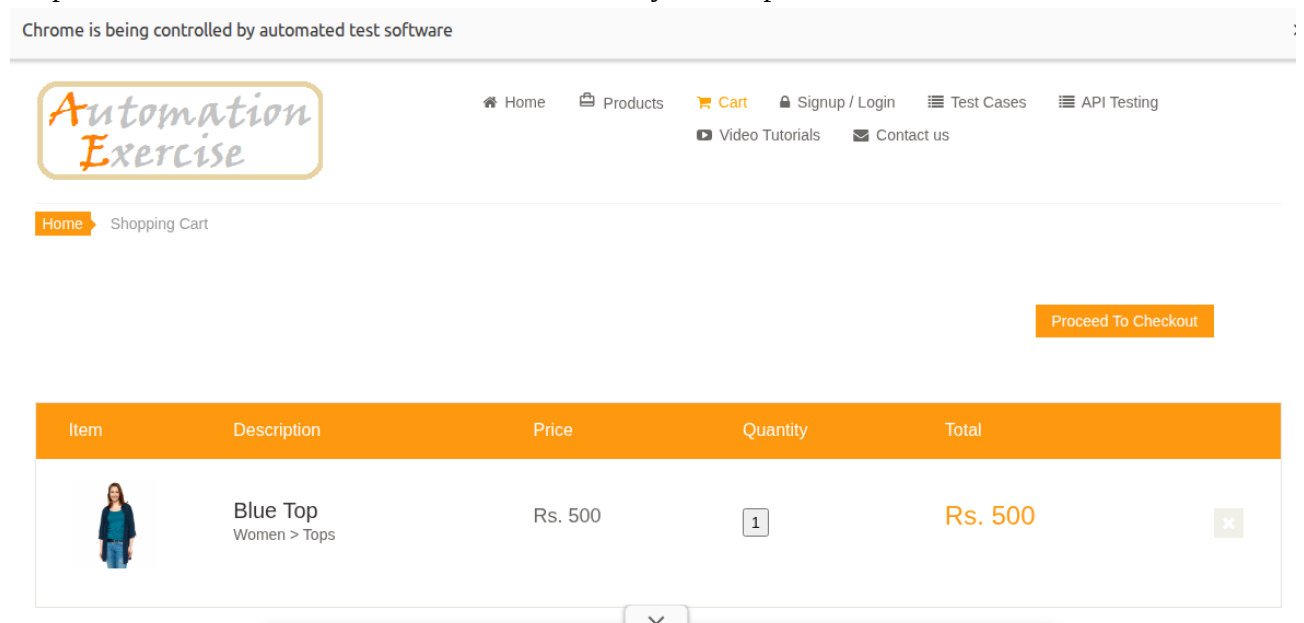
    } catch (Exception e) {
        System.out.println("TEST FAILED: " + e.getMessage());
    }

    System.out.println("Browser will remain open for inspection.");
}
}

```

#### Step 4: build and run the maven application

To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"** it opens automationexercise.com and automatically adds a product to cart



#### Step 5: now open the terminal and type the command **cd MyMavenExerciseAutomationApp**

to enter the folder and create a **Jenkinsfile** .Paste the code in the Jenkinsfile pipeline {

```
agent any // Use any available agent
```

```
tools {
```

```
    maven 'Maven' // Ensure this matches the name configured in Jenkins
```

```
}
```

```
stages {
```

```
    stage('Checkout') {
```

```
        steps {
```

```
            git
```

```
            branch:
```

```
            'master',
```

```
            url:
```

```
'https://github.com/Samiksha201104/MyMavenExerciseAutomationApp.git'
```

```

    }
}
stage('Build') {
    steps {
        sh 'mvn clean package' // Run Maven build
    }
}
stage('Test') {
    steps {
        sh 'mvn test' // Run unit tests
    }
}
stage('Run Application') {
    steps {
        // Start the JAR application
        sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
    }
}
}
post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}
}

```

Then commit and push it to GitHub

**Step 6:** in the Jenkins Dashboard click on new item type the name of the item and select Pipeline

## New Item

Enter an item name

MyMavenExerciseAutomationApp

Select an item type



### Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



### Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



### Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



### Multi-configuration project

OK

In configure scroll down to Pipeline and choose pipeline script from SCM, give the repository url and choose GitHub Credentials

Jenkins / MyMavenExerciseAutoma... / Configure

**Configure**

- General
- Triggers
- Pipeline**
- Advanced

**Definition**

Pipeline script from SCM

**SCM**

Git

**Repositories**

**Repository URL**

https://github.com/Samiksha201104/MyMavenExerciseAutomationApp.git

**Credentials**

GitHub Credentials

+ Add

Advanced

Save Apply

**Step 7 :** in the Dashboard click on the MyMavenExerciseAutomationApp item and select Build Now and then Full Stage view



## MyMavenExerciseAutomationApp - Stage View

|                                                     |                              | Declarative:<br>Checkout SCM | Declarative:<br>Tool Install | Checkout | Build        | Run<br>Application | Declarative:<br>Post Actions |
|-----------------------------------------------------|------------------------------|------------------------------|------------------------------|----------|--------------|--------------------|------------------------------|
| Average stage times:<br>(full run time: ~10min 39s) |                              | 1s                           | 229ms                        | 1s       | 3min 2s      | 40s                | 1s                           |
| #5                                                  | Apr 12<br>19:55<br>2 commits | 2s                           | 327ms                        | 2s       | 7min 59s     | 1min 54s           | 3s                           |
| #4                                                  | Apr 12<br>19:48<br>1 commit  | 1s                           | 241ms                        | 2s       | 59s          | 4s<br>failed       | 144ms                        |
| #3                                                  | Apr 12<br>19:44<br>3 commits | 1s                           | 120ms                        | 1s       | 7s<br>failed | 1s<br>failed       | 177ms                        |